

SENG 31242

SYSTEM DESIGN PROJECT



Group – Data Dynamos

SE/2022/001

SE/2022/006

SE/2022/018

SE/2022/036

SE/2022/045

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THE PROBLEM & PAIN POINTS

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Here are the problems our customers are facing

- Civil engineering firms lack an accessible, domain-specific tool to track daily task progress and asset usage against the original Bill of Quantities (BOQ)
- Current manual reporting creates delays
- Generic tools cannot calculate dynamic, real-time percentages of task completion or track the financial worth of assets used over specific time periods

TARGET USERS & **STAKEHOLDERS**

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QS Team / Managers (Web Users):

Responsible for creating the BOQ, inputting all tasks, and monitoring real-time progress and asset usage to date.

Field Supervisors (Mobile Users):

On-site workers whose primary role is to submit daily progress reports and log asset usage directly from the construction site.

EXISTING SOLUTIONS & **THEIR GAPS**

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- **Primavera P6:**
 - Very powerful, but too expensive, complex, and clunky for quick daily updates by field workers.
- **Generic Task Managers (e.g., Trello, Asana):**
 - "Simple to-do apps" that completely lack the specific business logic required for civil engineering, such as BOQ tracking and asset worth calculations.
- **Spreadsheets (Excel):**
 - Prone to data entry errors, need a specific person to maintain the logics manually, and hard for supervisors to update efficiently from mobile devices in the field.

PROPOSED SOLUTION

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A **customized, dual-platform**(Web + Mobile) **task management system** that acts as a streamlined "Mini Primavera P6," allowing field supervisors to **easily report daily progress** while giving QA teams a web-based dashboard to **track BOQ completion** and **asset usage** in **real-time**

CLIENT EVIDENCE

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FEASIBILITY SNAPSHOT

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Technical Feasibility:

- Our team possesses the skills to build a dual-platform system (React for web, Flutter for mobile, and Node.js for the Backend).

Timeline Feasibility:

- The scope is structured to fit an 8-week design phase and 4-month development window by strictly separating core tracking features from "nice-to-have" extras

UNIQUE VALUE PROPOSITION

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This Project Offers:

- Solution to a real world problem
- Complex business logic (calculating task percentages and asset worth against BOQs)
- User roll management
- Dual-platform architecture (web and mobile)

RISKS & MITIGATIONS

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- **Risk 1:** Heavy Dual-Platform Workload. Building both a web and mobile app in 4 months is ambitious. Mitigation: We will prioritize the web-based management panel and backend API. If we fall behind, we will delay the native mobile app and use a responsive web view for field workers.
- **Risk 2:** Complex BOQ Logic. Misunderstanding the customer's formulas could ruin the system. Mitigation: We will conduct extensive document analysis on the customer's existing BOQs during the Weeks 3-4 Requirements Engineering phase to lock down all calculations early.
- **Risk 3:** Client Availability. Mitigation: Schedule brief, mandatory sprint reviews to ensure we don't build the wrong features

Q&A